

KABA HE, China

WMO: 510530

Lat: 48.0516N

Lon: 86.4057E

Elev: 1752

StdP: 13.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.5	(c) -16.6	(d) -28.4	(e) 1.2	(f) -22.3	(g) -22.1	(h) 1.7	(i) -16.3	(j) 26.7	(k) 13.6	(l) 24.3	(m) 15.5	(n) 3.0	(o) 290	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.9	(c) 90.9	(d) 66.1	(e) 88.1	(f) 65.1	(g) 85.4	(h) 64.1	(i) 68.9	(j) 85.4	(k) 67.4	(l) 83.3	(m) 66.0	(n) 81.4	(o) 4.5	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 63.4	(b) 93.1	(c) 75.4	(d) 61.6	(e) 87.4	(f) 74.0	(g) 60.1	(h) 82.8	(i) 73.3	(j) 34.1	(k) 85.8	(l) 32.9	(m) 83.0	(n) 31.8	(o) 81.7	(p) 78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	19.5	17.3	-23.9	96.6	6.9	3.4	-28.8	99.0	-32.9	101.0	-36.8	102.8	-41.8	105.3
			-24.1	71.9	6.8	3.2	-29.1	74.2	-33.0	76.1	-36.9	77.9	-41.8	80.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.8	5.6	11.9	27.7	48.3	60.1	70.3	73.3	70.2	59.3	45.8	27.9	11.9
	DBStd	25.66	11.86	11.46	12.85	10.10	8.04	6.36	5.29	6.07	7.90	8.98	12.56	10.69
	HDD50	5343	1377	1066	693	146	14	0	0	0	15	187	665	1180
	HDD65	8784	1842	1486	1155	503	193	25	4	21	201	595	1114	1645
	CDD50	2733	0	0	3	94	326	609	722	625	296	57	1	0
	CDD65	699	0	0	0	1	41	183	261	182	31	0	0	0
	CDH74	6589	0	0	0	32	431	1688	2381	1758	296	3	0	0
	CDH80	2263	0	0	0	3	100	582	896	621	61	0	0	0
	WSAvg	7.1	9.1	8.4	7.7	8.0	7.3	5.7	4.8	4.9	5.5	6.5	7.9	9.2
	PrecAvg	6.9	0.2	0.2	0.3	0.6	0.9	0.7	0.9	0.6	0.6	0.7	0.7	0.4
(15) Precipitation	PrecMax	11.6	0.8	0.9	0.9	1.4	2.7	2.7	2.4	2.4	1.9	1.6	1.9	1.0
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1	0.2	0.0
	PrecStd	2.2	0.2	0.2	0.2	0.4	0.7	0.7	0.6	0.6	0.4	0.4	0.4	0.2
	PrecStd	2.2	0.2	0.2	0.2	0.4	0.7	0.7	0.6	0.6	0.4	0.4	0.4	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	31.7	35.5	61.8	77.9	87.4	92.7	95.8	93.7	86.4	72.8	56.3	35.2	
	0.4% MCWB	28.1	28.9	45.5	54.2	60.9	66.8	69.3	66.6	61.3	50.5	42.6	28.6	
	2% DB	26.1	31.5	55.5	72.9	82.9	89.2	91.2	89.8	80.9	67.3	51.4	31.5	
	2% MCWB	22.8	27.1	43.2	52.0	58.5	65.4	67.1	65.2	58.7	48.3	39.5	27.1	
	5% DB	23.0	28.3	50.2	68.4	79.2	86.2	88.1	86.4	77.1	63.1	47.3	27.9	
	5% MCWB	19.8	24.3	40.3	49.8	57.1	64.2	66.2	64.4	56.6	47.3	37.8	24.5	
	10% DB	20.0	25.3	44.7	64.3	75.4	83.2	85.1	83.4	73.5	59.2	43.2	24.9	
	10% MCWB	17.3	21.9	37.6	47.9	55.3	63.1	65.4	63.2	55.1	45.9	35.8	22.1	
	0.4% WB	28.6	30.1	47.3	56.3	63.3	69.6	72.6	70.1	63.1	53.2	45.4	30.8	
	0.4% MCDB	30.4	34.3	57.2	72.1	81.5	87.9	89.8	87.4	80.9	64.2	53.1	33.9	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	22.9	27.3	43.9	53.5	60.5	67.6	69.6	67.6	60.8	50.6	41.1	27.5	
	2% MCDB	25.6	31.2	54.4	67.9	77.3	84.7	85.8	84.6	77.6	62.3	48.4	30.8	
	5% WB	19.9	24.7	40.9	51.5	58.7	66.0	68.0	66.0	58.4	48.8	38.5	24.8	
	5% MCDB	22.6	28.1	49.4	65.0	74.7	82.1	83.5	82.7	73.5	60.3	45.8	27.8	
	10% WB	17.4	22.2	37.8	49.4	57.0	64.4	66.7	64.5	56.5	47.0	36.2	22.2	
	10% MCDB	20.0	25.0	44.6	61.4	72.3	80.0	81.5	80.0	70.2	57.8	42.9	24.7	
(35) Mean Daily Temperature Range	MDBR	10.9	12.6	15.1	19.8	20.7	20.4	19.9	21.4	20.7	16.3	11.0	9.5	
	5% DB	10.7	13.8	20.6	24.9	24.9	23.8	23.6	24.9	24.9	21.9	15.5	9.4	
	5% WB	8.6	10.8	11.8	11.3	9.9	9.6	9.7	10.4	11.0	10.5	8.9	7.2	
	MCDBR	10.7	13.2	19.2	22.3	21.6	21.0	20.7	22.3	22.3	19.5	14.5	9.0	
	MCWBR	9.0	10.4	11.5	10.9	9.3	9.0	8.8	10.0	10.7	10.0	8.9	7.2	
	MCWBR	9.0	10.4	11.5	10.9	9.3	9.0	8.8	10.0	10.7	10.0	8.9	7.2	
(40) Clear-Sky Solar Irradiance	taub	0.292	0.321	0.365	0.409	0.394	0.401	0.404	0.380	0.344	0.341	0.318	0.294	
	taud	2.114	2.073	2.073	2.070	2.176	2.236	2.252	2.289	2.340	2.248	2.163	2.093	
	Ebn at Noon	238	258	265	265	275	273	271	272	272	250	225	218	
	Edh at Noon	28	36	43	48	45	43	41	38	33	30	26	25	
(44) All-Sky Solar Radiation	RadAvg	547	884	1340	1704	2074	2163	2071	1879	1464	872	497	407	
	RadStd	36	37	73	98	115	118	124	81	109	86	52	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page